

Certificate of Analysis

Page 1 of 2

Client:	Orini Honey Packers Limited	Lab No:	4023878	HGPv1
Contact:	Mala Hansen C/- Orini Honey Packers Limited PO Box 12296 Chartwell Hamilton 3248	Date Received:	05-Nov-2025	
		Date Reported:	07-Nov-2025	
		Quote No:	102565	
		Order No:		
		Client Reference:	Bene Diastase	
		Submitted By:	Mala Hansen	

Sample Type: Honey

Sample Name:	OHP0725175	OHP0725176	OHP0225136
Lab Number:	4023878.1	4023878.2	4023878.3
Manuka Honey Analysis			
Dihydroxyacetone (DHA) mg/kg	559	-	-
5-Hydroxymethylfurfural (HMF) mg/kg	13.6	-	-
Methylglyoxal (MGO) mg/kg	125	-	-
Non Peroxide Activity (NPA)* % Phenol Equivalent	6.4	-	-
Diastase Analysis - Phadebas Method			
Diastase Activity DN	15.2	11.0	8.0

Summary of Methods

The following table(s) gives a brief description of the methods used to conduct the analyses for this job. The detection limits given below are those attainable in a relatively simple matrix. Detection limits may be higher for individual samples should insufficient sample be available, or if the matrix requires that dilutions be performed during analysis. A detection limit range indicates the lowest and highest detection limits in the associated suite of analytes. A full listing of compounds and detection limits are available from the laboratory upon request. Unless otherwise indicated, analyses were performed at Hill Labs, 28 Duke Street, Frankton, Hamilton 3204.

Sample Type: Honey

Test	Method Description	Default Detection Limit	Sample No
Diastase Activity	Aqueous extraction, analysed using the Phadebas amylase method. Diastase activity is expressed as the diastase number, equivalent to Schade units.	2.0 DN	1-3
3-in-1 Honey method	Aqueous extraction, derivatisation. Analysis by uHPLC / UV-Vis (dihydroxyacetone, 5-hydroxymethylfurfural, methylglyoxal). In-house.	1.0 - 10 mg/kg	1
Non Peroxide Activity (NPA)*	NPA is calculated from methylglyoxal using an industry accepted correlation curve based on published data ^{1,2} for NPA and the primary active ingredient, methylglyoxal. ¹ Isolation by HPLC and characterisation of the bioactive fraction of New Zealand manuka (<i>Leptospermum scoparium</i>) honey. C. J. Adams, et al. Carbohydrate Research 343 (2008) 651-659. ² Corrigendum to "Isolation by HPLC and characterization of the bioactive fraction of New Zealand manuka (<i>Leptospermum scoparium</i>) honey" [Carbohydr. Res. 343 (2008) 651]. C. J. Adams, et al. Carbohydrate Research 344 (2009) 2609.	1.0 % Phenol Equivalent	1

These samples were collected by yourselves (or your agent) and analysed as received at the laboratory.

Testing was completed between 05-Nov-2025 and 07-Nov-2025. For completion dates of individual analyses please contact the laboratory.

Samples are held at the laboratory after reporting for a length of time based on the stability of the samples and analytes being tested (considering any preservation used), and the storage space available. Once the storage period is completed, the samples are discarded unless otherwise agreed with the customer. Extended storage times may incur additional charges.

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